BioPrep

Product write-up for NEET (UG) Biology test prep · for owners and builders · 23 Oct 2026

What it is. A web app where students take timed NEET Biology mock tests, then learn from their mistakes with a tutor that **asks guiding questions before it reveals the answer**. The answer, written explanation and video unlock only after the student has reasoned it through, the 2:00 timer ends, or they tap Show me the answer.

How it earns. A free plan gives **2 full explanations in total**. After that, **Rs 100 per test** unlocks every solution and video in that test, permanently.

Who uses it

PERSON	WHAT THEY DO
Student (many under 18)	Signs in with a phone number and code, takes tests, reads results, opens solutions, pays to unlock, can download or delete their data.
Parent / guardian	No login. Gives consent during a minor's sign-up (name, mobile, relationship, a tick).
Teacher	Builds the question bank (typed or CSV), diagrams, videos and tests; sees content analytics.
Admin	Everything a teacher can, plus users and roles, payments, refund flags, revenue, manual unlocks.

A student's journey

- **1. Sign up.** Enter a mobile number, then the SMS code. First time only: name, NEET year, date of birth, accept the privacy policy. Under 18: a parent or guardian completes a consent step. Under 13 is refused.
- **2. Take a test.** Pick a Biology test, read the rules, start. One question at a time with a question palette, mark for review, and a visible timer. It submits itself at 00:00. Marking is +4, -1, 0.
- **3. Read the result.** Score, accuracy, correct / incorrect / skipped, Botany vs Zoology, chapter bars, time per question, and a review table. The correct answer stays hidden until the student opens that explanation.
- **4. Learn from a mistake.** Tap Explain. The tutor asks 2 or 3 short questions (tap an option or type). Up to 3 hints, each stronger. At the end: the answer, explanation, why each wrong option is wrong, key idea, common mistake, and a video.
- **5. Hit the paywall.** After 2 free explanations a locked screen offers Unlock for Rs 100. Pay by UPI or card on Razorpay. The test unlocks when the server receives Razorpay's confirmation, usually within seconds.
- **6. Stay in control.** Payment history, a revision list, English or Hindi, dark mode, privacy policy, Download my data, Delete my account.

Features at a glance

Tests

As a student I want an exam-like test.

- Timer with warnings, palette, mark for review, clear response
- Auto-submit at 00:00; answers kept if the network drops
- Scored on the server, never in the browser
- Option labels per question: A B C D, 1 2 3 4, i ii iii iv, Hindi letters

Results

As a student I want to know what to revise.

- Score, accuracy, attempted, time used
- Outcome donut with icons and text, never colour alone
- Subject and chapter bars; slow questions highlighted

Socratic tutor

As a student I want to work it out myself.

- Claude when configured, otherwise a built-in scripted guide
- 3-level hint ladder; 2:00 timer that pauses while thinking
- Never states the answer early; replies are checked
- Up to 5 messages per question

Video

As a student I want to watch the concept.

- Locked until the tutor session ends (paid: Skip to video)
- Private storage, short-lived links, speeds 1x / 1.5x / 2x
- Admin uploads MP4 up to 200 MB with a progress bar

Free plan and payments

As a student I want to try before I pay.

- 2 free explanations in total; one unlock = one whole test
- Razorpay UPI and cards; success shown only after the verified webhook
- Payment history; refund flagged by admin, refunded in Razorpay, test re-locks automatically

Privacy and children

As a parent I want my child protected.

- Only birth year stored; tutor chats never stored
- Guardian consent record for under 18
- Download or delete all data; old records purged on a schedule
- Public privacy policy page

Teacher tools

As a teacher I want to build content fast.

- Question form: explanation, guiding questions, diagram (alt text required), video
- CSV import with preview and per-row errors; rows land as drafts
- Test builder with publish switch

Admin tools

As an admin I want to run the business.

- Real analytics: average score per test, hardest questions, time per question, revenue
- Users with masked phones, role changes, manual unlock
- Payments list and refund flags

Rules that must not change without the owner's approval

RULE	VALUE
Live subjects	Biology (Botany and Zoology). Physics and Chemistry show as Coming soon.
Marking	+4 correct, -1 wrong, 0 skipped. Same for every question.
Free plan / price	2 explanations in total / Rs 100 per test, one time.
Tutor	2:00 per question, 3 hints, ends after 5 student messages. Video unlocks after the session ends, a timeout, or Show me the answer.
Roles	student, teacher, admin. Enforced on the server, never only on screen.

For builders: what you must understand

The one idea: the browser may read what a student is allowed to see. Anything secret or privileged goes through the API or a SQL function that checks who is asking. Never trust the client for scoring, price, entitlement or the timer.

PART	TECHNOLOGY AND ROLE
Website	React 18 + Vite (`src/`), deployed on Vercel. One store with the same action names for two backends: an offline demo (localStorage) and the live Supabase backend.
Database and login	Supabase Postgres with row-level security, phone-OTP sign-in, private *videos* bucket and public *diagrams* bucket (`supabase/migrations`).

API	Node + Express on Render (`server/`): tutor, explanations, video links, payments webhook, export and delete my data.
Shared code	`shared/`: product constants, tutor prompts and validation, option labels. Used by browser, API and tests.
Services	Claude API (tutor), Razorpay (payments), an SMS provider (OTP).

Security rules builders must keep

- **Secrets are in tables students cannot read.** Correct answers, explanations, video paths and tutor prompts live in `question_secrets` and `socratic_steps`, because row-level security cannot hide single columns.
- **One entitlement gate.** `private.open_explanation` decides staff preview, paid unlock, free slot or paywall. Clients have no write access to unlocks, free uses, sessions or usage. A lock stops parallel requests exceeding the free limit.
- **Only the Razorpay webhook unlocks.** Signature is checked over the raw body, amount matched to our own order, repeated events ignored. The browser's payment-success callback never unlocks anything. Price always comes from the database.
- **Tutor never leaks.** Prompts are built on the server; the browser receives only `{verdict, reply, step, established, choices, done}`. The answer and explanation come from a separate call that returns 403 until the server has marked the session finished.
- **Keys.** Anything named `VITE_*` is public. The service key, database URL, Anthropic key and Razorpay secrets live only on the API host.
- **Minimal data.** No chat text stored, birth year only, masked phones in admin, scheduled purge of old rows.

Working in the code

- **Run it:** `npm install`, `npm run dev:stack` (local database, sign-in stand-in, storage and API), then `npm run dev`. Test numbers: 9999999999 (admin), 9888888888 (teacher), code 123456.
- **Test it:** `npm test` (about 100 tests on a real Postgres engine: security rules, scoring, free and paid rules, payments, privacy) and `npm run test:e2e` (browser tests on desktop and a 390px phone).
- **Change the database** only with a new file in `supabase/migrations` once anything is deployed; then run `npm run gen:setup-sql`.
- **Conventions:** colours only from design tokens, light and dark; 44px touch targets; correct and incorrect always shown with an icon and text; student-facing text through `t()` (English first); admin numbers must be real.

Status, limits and next steps

Status: everything above is built and tested locally. Nothing has been run against real Supabase, Razorpay, Anthropic, Vercel or Render accounts yet. Follow `docs/LAUNCH.md` and do a test-mode payment and refund first.

- **Known limits:** a page refresh during a test loses the answers; video upload is a single request (a dropped connection restarts it); only the sign-in, header and home screens are translated; exactly four options per question; guardian consent is recorded, not verified.
- **Next:** go live; legal review of the privacy policy and consent method; native review of Hindi and a Meiteilon translation; resumable uploads and tests; Physics and Chemistry.

Full 21-page specification with all user stories and acceptance criteria: docs/spec/BioPrep-Product-Spec.pdf in the repository.